# ECON 6140 - Problem Set 5

## Question 1

[Bruce Hansen, *Econometrics*, Exercise 18.5] An economist is interested in the impact of Wisconsin's 2011 "Act 10" legislation on wages. (For background, Act 10 reduced the power of labor unions.) She computes the following statistics[1] for average wage rates in Wisconsin and the neighboring state of Minnesota for the decades before and after Act 10 was enacted.

| | Years | Average Wage |
|---|---|---|
| Wisconsin | 2001-2010 | 15.23 |
| Wisconsin | 2010-2020 | 16.72 |
| Minnesota | 2001-2010 | 16.42 |
| Minnesota | 2010-2020 | 18.10 |

a. Based on this information, what is her point estimate of the impact of Act 10 on average wages?

b. The numbers in the above table were calculated as county-level averages. (The economist was given the average wage in each county. She calculated the average for the state by taking the average across the counties.) Now suppose that she estimates the following linear regression, treating individual counties as observations.

$$wage = \alpha + \beta Act10 + \gamma Wisconsin + \delta Post2010 + e.$$

The three regressors are dummy variables for "Act 10 in effect in the state", "county is in Wisconsin", and "time period is 2011-2020." What value of $\widehat{\beta}$ does she find?

c. What value of $\widehat{\gamma}$ does she find?

## Question 2

Use the data in MROZ.raw to answer this question. The data consist of 753 observations. Before answering, remove 253 observations which will be used for an out-of-sample comparison.

a. Estimate $\mathrm{E}[inlf|X = \{kidslt6, kidsge6, age, educ, repwage, faminc, exper\}]$ using the OLS estimator. Calculate the Root Mean Square Error (RMSE) defined by

$$RMSE = \sqrt{\frac{1}{253}\sum_{i=1}^{253}(inlf_i - \widehat{inlf}_i)^2}$$

over the hold-out sample.

b. Estimate $\mathrm{E}[inlf|X = \{kidslt6, kidsge6, age, educ, repwage, faminc, exper\}]$ using LASSO with the penalty level chosen by cross-validation. Comment on which variables you chose. Calculate RMSE and compare with part a.

[1] These numbers are completely fictitious.

# Question 3

Use the data in headstart.dta to answer this question. These data come from Calonico, Cattaneo, Farrell, and Titiunik (2016) reanalysis of Ludwig, J., and Miller, D. L. (2007), "Does Head Start Improve Children's Life Chances? Evidence from a Regression Discontinuity Design". In this study, the unit of observation is the U.S. county, the treatment is receiving technical assistance to apply for Head Start funds, and the running variable is the county-level poverty index constructed in 1965. The RD design arises because the treatment was given only to counties whose poverty index was $x^* = 59.1984$ or above, a cutoff that was chosen to ensure that the 300 poorest counties received the treatment. The outcome of interest is the child mortality rate (for children of age five to nine) due to causes affected by Head Start's health services component. In the data set, the outcome variable is "mort_age59_related_postHS" ($y$) and the running variable is "povrate60" ($x$). There are also a few additional county-level demographic variables in the data.

a. Form a plot of $y$ against $x$ that provides visual evidence regarding the magnitude of discontinuity in $\mathrm{E}[Y|X = 59.1984]$. Form a similar plot with each of the county level demographic variables as the outcome. Does it appear that there is a discontinuity in the expected value of the county level demographics at $x^*$? Explain how looking at these additional plots provides a robustness check for the identifying assumption for the regression discontinuity design.

b. Estimate a linear model on both sides of the cutoff. Report the treatment effect estimator.